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BASIC MATHEMATICS

FORM FOUR

Certificate of Secondary Education Examination (CSEE) 2023 Answer Key and Worked Solutions

Subject	Basic Mathematics
Code	041
Level	Form Four
Year	2023
Exam Board	NECTA
Exam	Certificate of Secondary Education Examination (CSEE)
Type	Answer key and worked solutions
Questions	14 compulsory questions; 39 answered parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: BasicMath-F4-2023.pdf

Section A — Worked solutions

1(a)(i). Arrange 0.6, $\frac{3}{5}$, and 20% of $\frac{13}{4}$ in ascending order.

Step 1: $0.6 = 0.6666\dots$; $\frac{3}{5} = 0.6$; $(20/100) \times (\frac{13}{4}) = \frac{13}{20} = 0.65$.

Step 2: Compare decimal values: $0.6 < 0.65 < 0.6666\dots$.

Final Answer: $\frac{3}{5} < 20\% \text{ of } \frac{13}{4} < 0.6$.

Study Tip: The dot above 6 indicates a repeating decimal; it is not the terminating decimal 0.6.

1(a)(ii). Evaluate $13 - 2 \times 3 + 14 \div (2 + 5)$.

Step 1: Bracket first: $2 + 5 = 7$. Then multiplication and division: $2 \times 3 = 6$ and $14/7 = 2$.

Step 2: Now $13 - 6 + 2 = 9$.

Final Answer: 9.

1(b). Find the LCM of 2,3,5 by listing.

Step 1: Multiples of 2: 2,4,6,...,30; of 3: 3,6,9,...,30; of 5: 5,10,15,20,25,30.

Step 2: The first positive value in all three lists is 30.

Final Answer: 30.

2(a). Solve $5^{(x - 2y)} = 25$ and $3^{(2x)} \div 3^y = 3^4$.

Step 1: The first equation gives $x - 2y = 2$ because $25 = 5^2$. The quotient in the second gives $2x - y = 4$.

Step 2: Double the first: $2x - 4y = 4$. Subtract from the second to obtain $3y = 0$, so $y = 0$.

Step 3: Then $x = 2$. Check $5^2 = 25$ and $3^4/3^0 = 3^4$.

Final Answer: $x = 2$; $y = 0$.

2(b). Solve $4 + 3 \log_3 x = \log_3 24$.

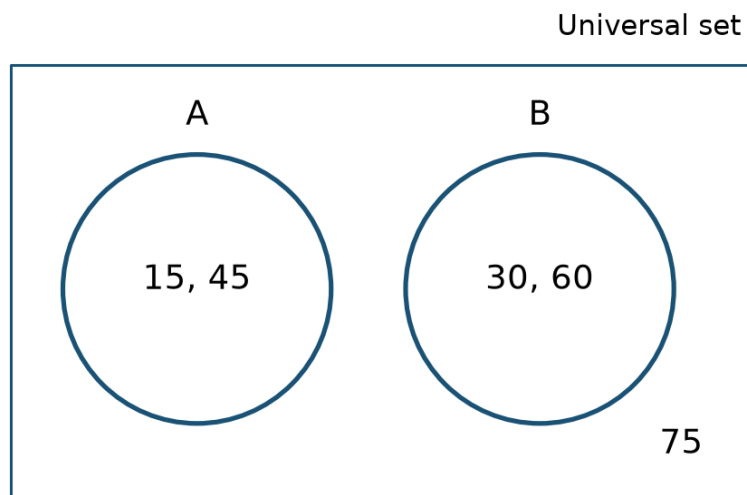
Step 1: For $x > 0$, write $4 = \log_3 81$ and $3 \log_3 x = \log_3 x^3$.

Step 2: Thus $\log_3(81x^3) = \log_3 24$, giving $x^3 = 24/81 = 8/27$.

Step 3: The positive cube root is $x = 2/3$.

Final Answer: $x = 2/3$.

3(a). Universal set = $\{15, 30, 45, 60, 75\}$, $A = \{15, 45\}$, $B = \{30, 60\}$. Find $(A \cup B)'$ and draw a Venn diagram.



Step 1: $A \cup B = \{15, 30, 45, 60\}$.

Step 2: The complement consists of universal elements not in that union: $\{75\}$.

Final Answer: $(A \cup B)' = \{75\}$. A and B are disjoint; put 75 outside both circles, within the universal set.

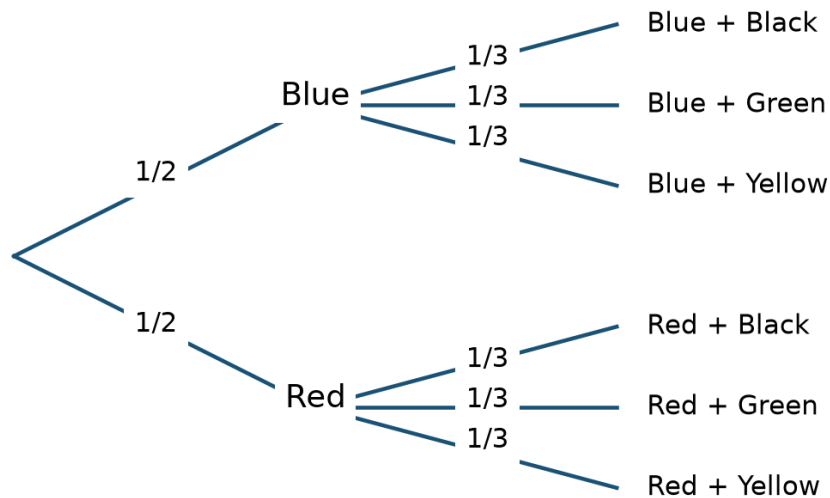
3(b)(i). Of 50 students, 35 are boys and 15 girls. Find the probability of choosing a boy randomly.

Step 1: Favourable outcomes = 35 boys; total equally likely outcomes = 50 students.

Step 2: Probability = $35/50 = 7/10$.

Final Answer: $7/10$.

3(b)(ii). Jonika has two shirts (blue, red) and three trousers (black, green, yellow). Use a tree diagram to find $P(\text{blue shirt and black trousers})$.



Step 1: First branch to blue or red, each probability $1/2$. From each, branch to black, green or yellow trousers, each probability $1/3$.

Step 2: Only one of the six equally likely outfits is blue-black. Multiply branch probabilities: $(1/2)(1/3) = 1/6$.

Final Answer: $1/6$, assuming equally likely independent choices.

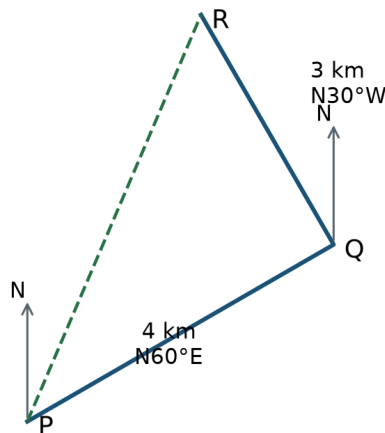
4(a). Show that $A(4, -4)$, $B(-6, -2)$, $C(2, 6)$ form an isosceles triangle.

Step 1: $AB^2 = (-6 - 4)^2 + (-2 + 4)^2 = 100 + 4 = 104$.

Step 2: $AC^2 = (2 - 4)^2 + (6 + 4)^2 = 4 + 100 = 104$. Equal squares imply equal positive lengths.

Final Answer: $AB = AC = \sqrt{104} = 2\sqrt{26}$, so the triangle is isosceles with vertex A.

4(b)(i). A man walks $PQ = 4$ km on bearing $N60^\circ E$, then $QR = 3$ km on $N30^\circ W$. Draw a labelled diagram.



Step 1: Use north-up axes. From P draw PQ 60° clockwise from north.

Step 2: From Q draw QR 30° west of north and label it 3 km. Join P to R for resultant displacement.

Final Answer: The diagram shows a 4 km north-east leg and a 3 km north-west leg, with north directions at P and Q.

4(b)(ii). Find resultant displacement from P to R.

Step 1: The leg directions are perpendicular. Thus resultant length $= \sqrt{4^2 + 3^2} = 5$ km.

Step 2: East component $= 4 \sin 60^\circ - 3 \sin 30^\circ = 2\sqrt{3} - 1.5 \approx 1.9641$ km; north component $= 4 \cos 60^\circ + 3 \cos 30^\circ = 2 + 1.5\sqrt{3} \approx 4.5981$ km.

Step 3: Angle east of north $= \arctan(1.9641/4.5981) \approx 23.13^\circ$. Displacement must include direction.

Final Answer: 5 km, direction $N23.13^\circ E$ (bearing approximately 023.13°).

5(a). $\angle RPQ = \angle PQR = 72^\circ$ and $\angle SPQ = \angle SQP$; $\angle PSQ = 112^\circ$. Find $\angle RPS$.

Step 1: In lower isosceles triangle PSQ, base angles $= (180 - 112)/2 = 34^\circ$.

Step 2: At P, $\angle RPS$ spans $\angle RPQ$ plus $\angle QPS$: $72 + 34 = 106^\circ$.

Final Answer: 106° .

5(b). A rectangle 72 m by 40 m and a triangle with base 60 m have equal area. Find the triangle height.

Step 1: Rectangle area = $72 \times 40 = 2880 \text{ m}^2$.

Step 2: $(1/2) \times 60 \times h = 2880$, so $30h = 2880$ and $h = 96 \text{ m}$.

Final Answer: 96 m.

6(a). Anna walks 24 km each day. Find her distance in 2 days in metres.

Step 1: Distance = $24 \times 2 = 48 \text{ km}$.

Step 2: Multiply by 1000 to convert kilometres into metres.

Final Answer: 48,000 m.

6(b)(i). Buying price C is proportional to selling price s; when $s = 20,000$, $C = 18,000$. Give the equation.

Step 1: Write $C = ks$, then $k = 18,000/20,000 = 0.9$.

Step 2: Substitute the constant into the proportionality equation.

Final Answer: $C = 0.9s$.

6(b)(ii). Find the new selling price if buying price rises by 15%.

Step 1: New buying price = $18,000 \times 1.15 = 20,700$.

Step 2: Keep $C = 0.9s$, so $s = 20,700/0.9 = 23,000$.

Final Answer: Tsh 23,000.

7(a). Ally and Jane share Tsh 64,000 in ratio 3:5. Find the difference in their shares.

Step 1: Total parts = 8; one part = $64,000/8 = 8,000$.

Step 2: Shares are 24,000 and 40,000; difference = 16,000.

Final Answer: Tsh 16,000.

7(b). Extract Mrisho's trial balance as at 28 February 2022 from his cash account.

Step 1: Total purchases = $1,000,000 + 1,400,000 = 2,400,000$; sales = $1,200,000 + 800,000 = 2,000,000$.

Step 2: Expenses, purchases and remaining cash have debit balances; sales and capital have credit balances.

Step 3: Check debits $2,400,000 + 200,000 + 900,000 = 3,500,000$ and credits $1,500,000 + 2,000,000 = 3,500,000$.

Final Answer: Debits: Purchases 2,400,000; Transport 200,000; Cash 900,000. Credits: Capital 1,500,000; Sales 2,000,000. Each total = Tsh 3,500,000.

8(a). Write the first four terms of $a_n = n(2n - 1)$ and say whether it is an AP or GP.

Step 1: Substitute $n = 1, 2, 3, 4$: $1 \times 1, 2 \times 3, 3 \times 5, 4 \times 7$.

Step 2: Differences are 5, 9, 13, which are unequal; ratios $6, 15/6, 28/15$ are also unequal. There is neither constant difference nor constant ratio.

Final Answer: 1, 6, 15, 28; neither an AP nor a GP.

8(b). The first eleven AP terms sum to 517 and the first term is 7. Find the sum of the fourth and ninth terms.

Step 1: $S_{11} = (11/2)[2(7) + 10d] = 517$; divide by $11/2$ to obtain $14 + 10d = 94$, so $d = 8$.

Step 2: $T_4 = 7 + 3(8) = 31$; $T_9 = 7 + 8(8) = 71$; sum = $31 + 71 = 102$.

Final Answer: 102.

9(a). A rectangular plot is 40 m long with diagonal 50 m. Find its width.

Step 1: Pythagoras gives $\text{width}^2 = 50^2 - 40^2 = 2500 - 1600 = 900$.

Step 2: Take the positive square root: width = 30 m.

Final Answer: 30 m.

9(b)(i). A is acute and $13 \cos A - 5 = 0$. Find $\tan A$ without tables.

Step 1: $\cos A = 5/13$, so take adjacent = 5 and hypotenuse = 13.

Step 2: The opposite side is $\sqrt{13^2 - 5^2} = 12$. Hence $\tan A = \text{opposite/adjacent} = 12/5$.

Final Answer: 12/5.

9(b)(ii). $AB = 8$ m, $AC = 5$ m and $\angle BAC = 60^\circ$. Find BC .

Step 1: Cosine rule: $BC^2 = 8^2 + 5^2 - 2(8)(5)\cos 60^\circ$.

Step 2: $BC^2 = 64 + 25 - 80(1/2) = 49$; $BC = 7$ m.

Final Answer: 7 m.

10(a). Roots of $ax^2 + bx + c = 0$ are -3 and $1/3$; a, b, c are integers. Find a, b, c .

Step 1: Form factors $(x + 3)(3x - 1) = 0$.

Step 2: Expand to $3x^2 + 8x - 3 = 0$ and compare coefficients.

Final Answer: Simplest integer coefficients: $a = 3$, $b = 8$, $c = -3$. Any common nonzero integer multiple gives the same roots.

Study Tip: The roots alone do not uniquely fix a, b, c unless a normalisation is specified. The displayed answer uses coprime integer coefficients.

10(b). Solve $10 - x \leq 3(x + 10)$ for integer x ; state the first four values.

Step 1: Expand: $10 - x \leq 3x + 30$. Move terms: $-20 \leq 4x$.

Step 2: Divide by positive 4 without reversing the inequality: $x \geq -5$.

Final Answer: $x \geq -5$, $x \in \mathbb{Z}$; first four: $-5, -4, -3, -2$.

Section Summary: Inspect recurring-decimal dots and operation signs before calculating. Match every answer to the stated method and explain when coefficients are determined only up to a common factor.

Section B — Worked solutions

11(a). Classes 40-49,...,90-99 have frequencies 2,4,7,9,5,3. Find the median to 2 decimal places.

Step 1: Cumulative frequencies are 2,6,13,22,27,30. $N/2 = 15$ lies in 70-79.

Step 2: Use boundaries and interpolation: median = $69.5 + [(15 - 13)/9] \times 10 = 71.7222\dots$, so 71.72.

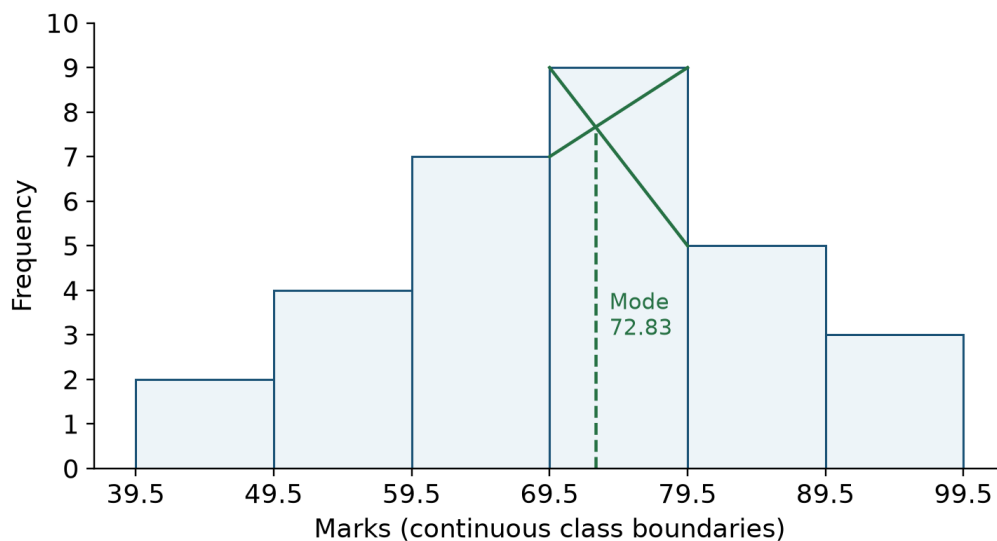
Final Answer: 71.72 marks.

11(b). Find the grouped mean to four significant figures.

Step 1: Midpoints are 44.5, 54.5, 64.5, 74.5, 84.5, 94.5. Products with frequencies are 89, 218, 451.5, 670.5, 422.5, 283.5; sum = 2135.

Step 2: Mean = $2135/30 = 71.1666\dots$. To four significant figures, retain 7, 1, 1, 6 and round the last 6 up to 7.

Final Answer: 71.17 marks.

11(c). Draw a histogram and estimate its mode.

Step 1: Use continuous class boundaries 39.5, 49.5, ..., 99.5 and adjacent bars of heights 2, 4, 7, 9, 5, 3.

Step 2: The modal class is 70–79. $d_1 = 9 - 7 = 2$, $d_2 = 9 - 5 = 4$.

Step 3: Mode = $69.5 + [2/(2 + 4)] \times 10 = 72.8333\dots$. The histogram includes the graphical interpolation.

Final Answer: Mode $\approx$ 72.83 marks.

12(a)(i). A cube has edge 8 cm. Find total surface area.

Step 1: Each of six faces has area $8^2 = 64 \text{ cm}^2$.

Step 2: Total area = $6 \times 64 = 384 \text{ cm}^2$.

Final Answer: 384 cm².

12(a)(ii). Find the angle between space diagonal AF and base plane ABCD, to the nearest degree.

Step 1: F projects onto C in the base. $AC = 8\sqrt{2}$ and $CF = 8$, so the line-plane angle is $\angle FAC$.

Step 2: $\tan \angle FAC = CF/AC = 1/\sqrt{2}$; $\angle FAC \approx 35.264^\circ$, rounding to 35° .

Final Answer: 35° .

12(b). A plane flies from $7^\circ\text{S}, 45^\circ\text{E}$ to $9^\circ\text{N}, 45^\circ\text{E}$ at 500 km/h, departing 8:00 a.m. Use 112 km per degree. Find arrival time.

Step 1: Latitude difference $= 7 + 9 = 16^\circ$. Distance $= 16 \times 112 = 1792$ km.

Step 2: Time $= 1792/500 = 3.584$ hours $= 3$ hours 35.04 minutes. Add to 8:00 a.m.

Final Answer: Approximately 11:35 a.m. the same day (exactly 11:35:02.4 a.m. in the stated model).

13(a). Solve $[[x, 4], [3, y]] + [[3x, 5], [-3, 7]] = [[8, 9], [0, 9]]$.

Step 1: Equate corresponding entries: $x + 3x = 8$ and $y + 7 = 9$.

Step 2: Thus $4x = 8$ and $x = 2$; $y = 2$. The other entries check: $4 + 5 = 9$ and $3 - 3 = 0$.

Final Answer: $x = 2$; $y = 2$.

13(b)(i). Anna answers 49 questions, scoring +2 per correct and -1 per incorrect for total 62. Represent in matrix form.

Step 1: Counts give $x + y = 49$. Scores give $2x - y = 62$.

Step 2: Put the coefficients of x and y in rows, and totals in the right-hand column vector.

Final Answer: $[[1, 1], [2, -1]][[x], [y]] = [[49], [62]]$.

13(b)(ii). Use the inverse matrix method to find the correct answers.

Step 1: $A = [[1, 1], [2, -1]]$ has determinant -3 . $A^{-1} = (1/-3)[[-1, -1], [-2, 1]]$.

Step 2: Multiply by $[[49], [62]]$: $(1/-3)[[-111], [-36]] = [[37], [12]]$. Check $2(37) - 12 = 62$.

Final Answer: 37 correct answers (and 12 incorrect).

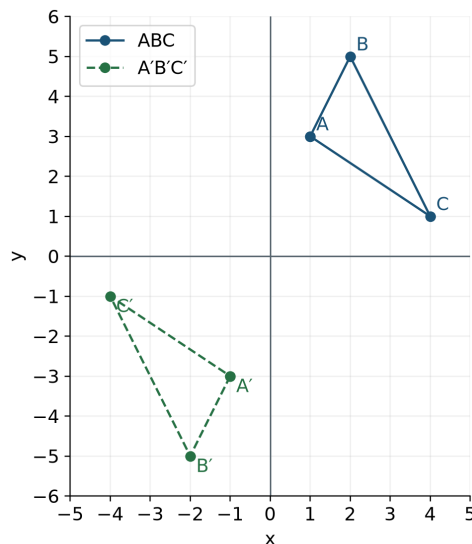
13(c)(i). Rotate $A(1,3), B(2,5), C(4,1)$ by 180° anticlockwise about the origin.

Step 1: A 180° rotation has matrix $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$.

Step 2: It maps (x,y) to $(-x,-y)$; change both signs of every vertex.

Final Answer: $A'(-1,-3), B'(-2,-5), C'(-4,-1)$.

13(c)(ii). Draw ABC and $A'B'C'$ on the same axes.



Step 1: Plot $A(1,3), B(2,5), C(4,1)$ and join them.

Step 2: Plot the three negative-coordinate images and join them, using a distinct line style. Use equal horizontal and vertical scales.

Final Answer: The labelled original triangle is in quadrant I and its 180° image is in quadrant III.

14(a)(i). For integer x , $f(x) = -2$ when $0 < x \leq 5$; $f(x) = x + 1$ when $-6 \leq x < 0$. Find $f(4)$, $f(-5)$.

Step 1: 4 lies in the positive branch, where the function is the constant -2 .

Step 2: -5 lies in the negative branch, so $f(-5) = -5 + 1 = -4$.

Final Answer: $f(4) = -2$; $f(-5) = -4$.

14(a)(iii). State the integer domain and range of that piecewise function.

Step 1: The two input intervals include the integers -6 through -1 and 1 through 5 ; zero is excluded.

Step 2: The negative branch outputs -5 through 0 ; the positive branch adds only -2 , already in that set.

Final Answer: Domain = $\{-6, -5, -4, -3, -2, -1, 1, 2, 3, 4, 5\}$. Range = $\{-5, -4, -3, -2, -1, 0\}$.

Study Tip: The paper labels this part (iii) after (i); the original label is retained.

14(b). Type A aircraft need 6 dam² parking and cost 20 billion; B need 2 dam² and cost 30 billion. Limits are 60 dam² and 480 billion. Find the greatest aircraft count.

Step 1: For nonnegative integer counts x, y , constraints are $3x + y \leq 30$ and $2x + 3y \leq 48$. Objective = $x + y$.

Step 2: Feasible vertices are $(0,0), (10,0), (6,12), (0,16)$, giving totals $0, 10, 18, 16$. The intersection is obtained from $3x + y = 30$ and $2x + 3y = 48$.

Step 3: At $(6,12)$, parking = $6(6) + 2(12) = 60$ and cost = $20(6) + 30(12) = 480$. The integer solution reaches both limits.

Step 4: For a direct optimality bound, add $1/7$ of the first inequality and $2/7$ of the second: $x + y \leq 18$.

Final Answer: 18 aircraft: 6 of type A and 12 of type B.

Section Summary: Use continuous class boundaries for grouped interpolation. Draw graphs with equal scales, respect integer domains, and test all relevant feasible vertices.